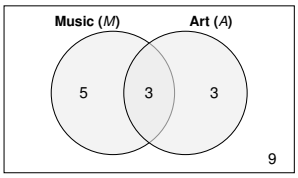
Unit 6 Review KEY

Name _____

One class, every rule. A class has 20 students. 8 are in the music club (M). 6 are in the art club (A). 3 are in both clubs.

6.1 Add for "or." Add the two groups. Then subtract the overlap, because it was counted twice.	 $P(M \cup A) = P(M) + P(A) - P(M \cap A)$ $= \frac{8}{20} + \frac{6}{20} - \frac{3}{20} = \frac{11}{20} = 0.55$ Complement: $P(\text{not } M) = 1 - \frac{8}{20} = \frac{12}{20} = 0.6$ **Not** mutually exclusive: 3 are in both.
6.2 Multiply for "and." Independent: $P(A) \cdot P(B)$. Dependent: use the new numbers after the first pick.	Pick a student on Monday and on Tuesday. The same student can be picked twice: **independent**. $P(M, M) = \frac{8}{20} \cdot \frac{8}{20} = 0.16$ $P(\text{at least one } M) = 1 - 0.6 \cdot 0.6 = 0.64$ Pick 2 **different** students: **dependent**. 7 of the 19 left are in music. $P(M, M) = \frac{8}{20} \cdot \frac{7}{19} = \frac{56}{380} = \frac{14}{95} \approx 0.15$
6.3 Look inside the group. $P(A \mid B)$ is the probability of A **given** B. Look only at the B group.	Look only at the 8 music students. 3 of them are in art. $P(A \mid M) = \frac{3}{8} = 0.375$ but $P(A) = \frac{6}{20} = 0.3$ They are different, so music and art are **not** independent.
6.4 Simulate. Trials give an experimental probability. More trials get closer to the theoretical probability.	$P(M) = 0.4$. Digits 0 to 3 = music (4 of the 10 digits). One trial = 2 digits (Monday, Tuesday). **91 74 [00] 29 75 50 47 36 88 [13]** 2 of the 10 trials have music both days. Experimental: $\frac{2}{10} = 0.2$ Theoretical: $0.4 \cdot 0.4 = 0.16$
6.5 Count arrangements. Order matters: a **permutation**. $n! = n \cdot (n-1) \cdots 2 \cdot 1$	The 8 music students choose a leader, a helper, and a note-taker. The jobs are different, so order matters. ${}_8P_3 = \frac{8!}{5!} = 8 \cdot 7 \cdot 6 = 336$ ways
6.6 Count groups. Order does not matter: a **combination**. Divide by the number of orders.	Choose 3 of the 8 music students to play at a school event. They all have the same job. ${}_8C_3 = \frac{8!}{3!5!} = \frac{336}{3!} = \frac{336}{6} = 56$ groups
6.7 Count to find a probability. Count the ways you want and all the ways. Use the same method for both.	The 3 players are chosen at random. What is the probability that all 3 are also in art? Only 3 music students are in art, so ${}_3C_3 = 1$ group works. $P(\text{all 3 in art}) = \frac{1}{56} \approx 0.02$

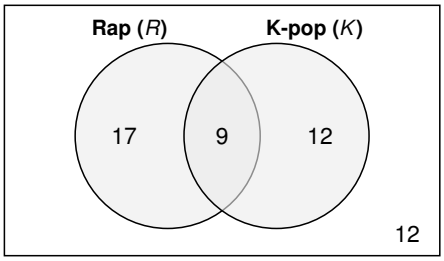
Does order matter? If I switch the order, is it a different result? Different jobs: permutation. Same job: combination.

Work with a partner. Compare answers when you finish each problem.

6.1 - Venn Diagrams and the Addition Rule

1. **I find a probability and its complement, and I read the symbols $A \cup B$ and $A \cap B$.**

   50 students said if they listen to rap (R) and if they listen to K-pop (K).



(a) Find $P(R)$.

$$\frac{17+9}{50} = \frac{26}{50} = \frac{13}{25} = 0.52$$

(b) Find $P(\text{not } R)$.

$$1 - \frac{13}{25} = \frac{12}{25} = 0.48$$

(c) Find $P(K)$. $\frac{9+12}{50} = \frac{21}{50} = 0.42$

(d) What does $R \cap K$ mean in words? Find $P(R \cap K)$.

Listens to rap **and** K-pop. $\frac{9}{50} = 0.18$

(e) What does $R \cup K$ mean in words? Find $P(R \cup K)$.

Listens to rap **or** K-pop **or** both. $\frac{17+9+12}{50} = \frac{38}{50} = \frac{19}{25} = 0.76$

(f) Find $P(\text{neither})$. $\frac{12}{50} = \frac{6}{25} = 0.24$

(g) Find $P(\text{rap only})$. $\frac{17}{50} = 0.34$

(h) Explain why $P(R \cup K) + P(\text{neither}) = 1$.

"Neither" is the complement of $R \cup K$. Every student is in exactly one of the two groups.
$0.76 + 0.24 = 1$

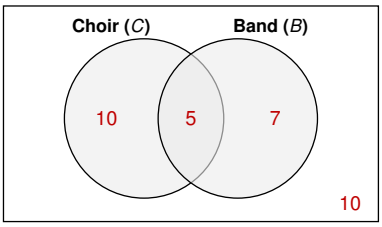
2. **I use the addition rule, and I decide if two events are mutually exclusive.**

(a) $P(A) = 0.45$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$. Find $P(A \cup B)$.
$P(A \cup B) = 0.45 + 0.3 - 0.1 = 0.65$

(b) A and B are mutually exclusive. $P(A) = 0.25$ and $P(B) = 0.35$. Find $P(A \cap B)$ and $P(A \cup B)$.
$P(A \cap B) = 0$, because they cannot both happen. $P(A \cup B) = 0.25 + 0.35 = 0.6$

In a class of 32 students, 15 take choir (C), 12 take band (B), and 5 take both.

(c) Fill in the Venn diagram.



(d) Use the addition rule to find $P(C \cup B)$.
$\frac{15}{32} + \frac{12}{32} - \frac{5}{32} = \frac{22}{32} = \frac{11}{16} \approx 0.69$

Roll a number cube (1 to 6). Are the events mutually exclusive? Find the probability.

(e) a 2 or a 5
Yes. One roll cannot be both. $\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$

(f) an even number or a number greater than 3
No. 4 and 6 are both. $\frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}$

Draw one card from a deck of 52. Are the events mutually exclusive? Find the probability.

(g) a diamond or a black card
Yes. Diamonds are red, so no card is both. $\frac{13}{52} + \frac{26}{52} = \frac{39}{52} = \frac{3}{4}$

(h) a red card or a queen
No. The queen of hearts and the queen of diamonds are both.
$\frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13} \approx 0.54$

6.2 - The Multiplication Rule

3. **I use the multiplication rule for independent events and the complement for "at least one" questions.**

   A soccer player scores on 75% of her penalty kicks. The kicks are independent. She takes 2 kicks.

   (a) Find P(scores both). $0.75 \cdot 0.75 = 0.5625$

   (b) Find P(misses both). $0.25 \cdot 0.25 = 0.0625$

   (c) Find P(scores at least one). $1 - P(\text{misses both}) = 1 - 0.0625 = 0.9375$

   Spin a spinner with 3 equal parts (red, blue, green). Then roll a number cube.

   (d) Find $P(\text{red} \cap 6)$. $\frac{1}{3} \cdot \frac{1}{6} = \frac{1}{18}$

   A quiz has 3 questions. Each question has 5 choices. You guess on all 3.

   (e) Find P(all 3 right). $\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{125} = 0.008$

   (f) Find P(at least one wrong). $1 - \frac{1}{125} = \frac{124}{125} = 0.992$

   (g) Find P(all 3 wrong). $\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} = \frac{64}{125} = 0.512$

   (h) Find P(at least one right). $1 - \frac{64}{125} = \frac{61}{125} = 0.488$

4. **I decide if two events are independent or dependent, and I use the general multiplication rule.**

A jar has 7 red beads and 3 blue beads. You take 2 beads without looking.

(a) You put the first bead back before you take the second. Find $P(\text{red, red})$.

Independent. $\frac{7}{10} \cdot \frac{7}{10} = \frac{49}{100} = 0.49$

(b) You keep the first bead out. Find $P(\text{red second} \mid \text{red first})$.

6 red beads are left out of 9 beads. $\frac{6}{9} = \frac{2}{3}$

(c) You keep the first bead out. Find $P(\text{red, red})$.

Dependent. $\frac{7}{10} \cdot \frac{6}{9} = \frac{42}{90} = \frac{7}{15} \approx 0.47$

(d) You keep the first bead out. Find $P(\text{blue, blue})$.

$\frac{3}{10} \cdot \frac{2}{9} = \frac{6}{90} = \frac{1}{15} \approx 0.07$

(e) You keep the first bead out. Find $P(\text{red first, then blue})$.

$\frac{7}{10} \cdot \frac{3}{9} = \frac{21}{90} = \frac{7}{30} \approx 0.23$

(f) $P(A) = 0.5$ and $P(B \mid A) = 0.3$. Find $P(A \cap B)$.

$P(A \cap B) = P(A) \cdot P(B \mid A) = 0.5 \cdot 0.3 = 0.15$

Independent or dependent? Explain.

(g) Roll a number cube two times.

Independent. The first roll does not change the second roll.

(h) Choose 2 different students from a class to read aloud.

Dependent. The first student cannot be chosen again, so the choices change.

6.3 - Conditional Probability

5. **I find conditional probabilities from a two-way table, and I check if two events are independent.**

120 students said if they have a library card and if they read for fun.

	Reads for fun	Does not	Total
Library card	36	24	60
No library card	18	42	60
Total	54	66	120

(a) Find $P(\text{reads})$. $\frac{54}{120} = \frac{9}{20} = 0.45$

(b) Find $P(\text{reads} \mid \text{library card})$. Only the 60 with a card. $\frac{36}{60} = \frac{3}{5} = 0.6$

(c) Find $P(\text{library card} \mid \text{reads})$. Only the 54 who read. $\frac{36}{54} = \frac{2}{3} \approx 0.67$

(d) Find $P(\text{reads} \mid \text{no library card})$. $\frac{18}{60} = \frac{3}{10} = 0.3$

(e) Find $P(\text{no library card} \mid \text{does not read})$. $\frac{42}{66} = \frac{7}{11} \approx 0.64$

(f) Are reading for fun and having a library card independent? Explain.
No. $P(\text{reads}) = 0.45$, but $P(\text{reads} \mid \text{library card}) = 0.6$. Knowing about the card changes the probability.

80 students said if they are left-handed and if they play an instrument.

	Instrument	No instrument	Total
Left-handed	6	14	20
Right-handed	18	42	60
Total	24	56	80

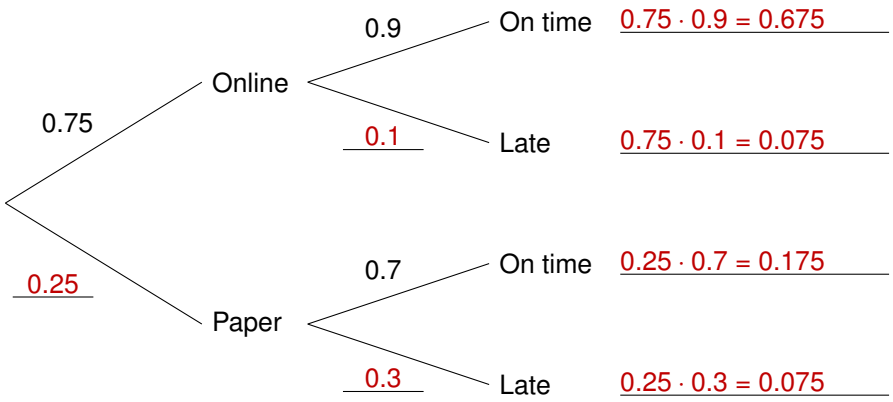
(g) Find $P(\text{instrument} \mid \text{left-handed})$. $\frac{6}{20} = \frac{3}{10} = 0.3$

(h) Are being left-handed and playing an instrument independent? Show your work.
Yes. $P(\text{instrument}) = \frac{24}{80} = 0.3$ and $P(\text{instrument} \mid \text{left-handed}) = 0.3$. They are equal.

6. **I use a tree diagram and the notation $P(A \mid B)$ to find probabilities.**

75% of students do their homework online. 25% do it on paper. 90% of online homework is turned in on time. 70% of paper homework is turned in on time.

(a) Fill in the tree diagram. Write the missing probabilities and multiply along each path.



(b) Find $P(\text{online} \cap \text{on time})$. $0.75 \cdot 0.9 = 0.675$

(c) Find $P(\text{on time})$. $0.675 + 0.175 = 0.85$

(d) Find $P(\text{late})$ two ways.
Add the late paths: $0.075 + 0.075 = 0.15$. Use the complement: $1 - 0.85 = 0.15$.

(e) The homework was on time. Find $P(\text{paper} \mid \text{on time})$.
$\frac{0.175}{0.85} = \frac{7}{34} \approx 0.21$

(f) The homework was late. Find $P(\text{online} \mid \text{late})$.
$\frac{0.075}{0.15} = 0.5$

(g) Find $P(\text{on time} \mid \text{paper})$. Where on the tree do you see it?
0.7. It is on the branch after Paper. Check: $\frac{0.175}{0.25} = 0.7$

6.4 - Simulations

7. **I tell theoretical and experimental probability apart, and I explain the law of large numbers.**

A number cube is rolled 60 times.

Number	1	2	3	4	5	6
Times rolled	8	12	9	7	10	14

(a) Find the experimental probability of a 6. $\frac{14}{60} = \frac{7}{30} \approx 0.23$

(b) Find the theoretical probability of a 6. $\frac{1}{6} \approx 0.17$

(c) Find the experimental probability of an even number. $\frac{12+7+14}{60} = \frac{33}{60} = \frac{11}{20} = 0.55$

(d) Find the theoretical probability of an even number. $\frac{3}{6} = \frac{1}{2} = 0.5$

(e) Why are the experimental and theoretical probabilities different?
Chance. 60 rolls is not very many trials.

(f) The cube is rolled 6,000 times. What number do you expect the experimental probability of a 6 to be close to? Why?
Close to $\frac{1}{6} \approx 0.17$.
The law of large numbers: with more trials, the experimental probability gets closer to the theoretical probability.

Theoretical or experimental? Explain.

(g) It rained on 12 of the last 30 days, so $P(\text{rain}) = \frac{12}{30} = 0.4$.
Experimental. It comes from what happened.

(h) A spinner has 8 equal parts, so the probability of each part is $\frac{1}{8}$.
Theoretical. It comes from reasoning about equally likely outcomes.

8. **I assign random digits and run a simulation to estimate a probability.**

A seed sprouts 80% of the time. Plant 3 seeds. Simulate:
Digits 0 to 7 = sprouts. Digits 8 and 9 = does not sprout. One trial = 3 digits (3 seeds).

(a) Why do the digits 0 to 7 stand for a seed that sprouts?
0 to 7 is 8 of the 10 digits, and $\frac{8}{10}$ = 80%.

(b) Use these random digits to run 10 trials. Count the seeds that sprout in each trial.

17118 89032 50415 68007 77585 15165

Trial	1	2	3	4	5	6	7	8	9	10
Digits	171	188	903	250	415	680	077	758	515	165
Sprouts	3	1	2	3	3	2	3	2	3	3

(c) Estimate P(all 3 sprout). Trials 1, 4, 5, 7, 9, and 10. $\frac{6}{10} = 0.6$

(d) Estimate P(exactly 2 sprout). Trials 3, 6, and 8. $\frac{3}{10} = 0.3$

(e) The theoretical probability that all 3 sprout is $0.8 \cdot 0.8 \cdot 0.8 = 0.512$. How could you make your estimate closer?
Run many more trials.

How would you assign random digits to simulate each event?

(f) A player makes 90% of her free throws. 0 to 8 = make, 9 = miss

(g) An event with $P = 0.15$. Use pairs: 00 to 14 = yes, 15 to 99 = no

(h) A spinner with 3 equal parts: red, blue, and green.
1, 2, 3 = red; 4, 5, 6 = blue; 7, 8, 9 = green; skip 0

6.5 - Permutations

9. **I use the counting principle and factorials.**

Find each value.

(a) 6! $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$

(b) $\frac{9!}{7!}$ $9 \cdot 8 = 72$

(c) 0! 1

(d) A video game character has 5 hair styles, 4 shirts, and 3 hats. How many different characters can you make?
$5 \cdot 4 \cdot 3 = 60$ characters

(e) A school ID has 2 letters (A to Z) and then 3 digits (0 to 9). Letters and digits can repeat. How many IDs are there?
$26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 676{,}000$ IDs

(f) In how many ways can 8 students line up for a photo?
$8! = 40{,}320$ ways

(g) In how many ways can you arrange the letters of the word PLANT?
$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways

10. **I count permutations with ${}_nP_r$, and I tell when repeats are allowed.**

(a) Find ${}_7P_2$. $7 \cdot 6 = 42$

(b) Find ${}_9P_4$. $9 \cdot 8 \cdot 7 \cdot 6 = 3{,}024$

(c) 11 students run for president, vice president, secretary, and treasurer. In how many ways can the 4 jobs be filled?
${}_{11}P_4 = 11 \cdot 10 \cdot 9 \cdot 8 = 7{,}920$ ways

(d) 7 swimmers are in a race. In how many ways can gold, silver, and bronze medals be given?
${}_7P_3 = 7 \cdot 6 \cdot 5 = 210$ ways

A bike lock code has 5 digits (0 to 9).

(e) How many codes are there if digits can repeat? $10^5 = 100{,}000$

(f) How many codes are there if no digit repeats? ${}_{10}P_5 = 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30{,}240$

(g) A code uses 3 of the vowels A, E, I, O, U. Repeats are allowed. How many codes are there?
$5 \cdot 5 \cdot 5 = 5^3 = 125$ codes

(h) Which answer is a permutation, (e) or (f)? Explain.
(f). No digit repeats, and order matters. With repeats, it is not a permutation. Use the counting principle.

6.6 - Combinations

11. **I count combinations with ${}_nC_r$.**

Find each value.

(a) ${}_7C_2$ $\quad \dfrac{7 \cdot 6}{2 \cdot 1} = 21$

(b) ${}_9C_3$ $\quad \dfrac{9 \cdot 8 \cdot 7}{3 \cdot 2 \cdot 1} = 84$

(c) ${}_{10}C_4$ $\quad \dfrac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$

(d) $\binom{6}{5}$ $\quad \dfrac{6!}{5!\,1!} = 6$

(e) 15 people meet. Each person shakes hands once with every other person. How many handshakes are there?

${}_{15}C_2 = \dfrac{15 \cdot 14}{2} = 105$ handshakes

(f) You choose 5 of 12 songs to download. In how many ways can you choose them?

${}_{12}C_5 = 792$ ways

(g) A salad bar has 8 toppings. A salad has 4 different toppings. How many different salads are there?

${}_8C_4 = \dfrac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70$ salads

12. **I decide whether a problem needs a permutation or a combination, and I explain the difference.**

For each situation, write permutation or combination. Then find the number of ways.

Situation	**Permutation or combination?**	**Number of ways**
(a) Choose 4 of 14 students to carry a banner.	combination	${}_{14}C_4 = 1{,}001$
(b) Give gold, silver, and bronze medals to 3 of 14 dancers.	permutation	${}_{14}P_3 = 2{,}184$
(c) Choose a captain and a co-captain from 16 players.	permutation	${}_{16}P_2 = 240$
(d) Choose 2 of 16 players to carry the water.	combination	${}_{16}C_2 = 120$
(e) Arrange 6 pictures in a row on a wall.	permutation	$6! = 720$
(f) Choose 3 of 13 books to read this summer.	combination	${}_{13}C_3 = 286$

(g) Why is the answer to (c) 2 times the answer to (d)?

Each pair of players can be put in order 2! = 2 ways. A permutation counts both orders. A combination counts the pair once.

(h) Which situation is a permutation? Circle the letter.

A Choosing 3 of 20 students to go on a field trip.

B Choosing 4 of 10 songs to download.

(C) Choosing the first, second, and third chair from 9 violin players.

D Choosing 2 of 6 colors to paint a room.

6.7 - Probability with Counting

13. **I count the possible outcomes and the outcomes I want, and I find the probability.**

A playlist has 10 songs: 6 rap songs and 4 K-pop songs. The app picks 3 songs at random. Order does not matter.

(a) How many groups of 3 songs are possible? ${}_{10}C_3 = 120$

(b) Find P(all 3 are rap). ${}_6C_3 = 20$. $\frac{20}{120} = \frac{1}{6} \approx 0.17$

(c) Find P(all 3 are K-pop). ${}_4C_3 = 4$. $\frac{4}{120} = \frac{1}{30} \approx 0.03$

(d) Find P(not all 3 are rap). Complement: $1 - \frac{1}{6} = \frac{5}{6} \approx 0.83$

A club has 13 members: 5 seniors and 8 juniors. 2 members are chosen at random to go to a meeting.

(e) How many pairs are possible? ${}_{13}C_2 = \frac{13 \cdot 12}{2} = 78$

(f) Find P(both seniors). ${}_5C_2 = 10$. $\frac{10}{78} = \frac{5}{39} \approx 0.13$

(g) Find P(both juniors). ${}_8C_2 = 28$. $\frac{28}{78} = \frac{14}{39} \approx 0.36$

(h) Find P(one senior and one junior).

$5 \cdot 8 = 40$ pairs. $\frac{40}{78} = \frac{20}{39} \approx 0.51$. Check: $10 + 28 + 40 = 78$.

14. **I decide if order matters in a probability problem.**

20 students are in a class. Does order matter? Find the probability.

(a) The teacher picks 2 students at random. One reads aloud and one writes on the board. What is the probability that Ana reads and Ben writes?

Yes, the jobs are different. ${}_{20}P_2 = 20 \cdot 19 = 380$. $P = \frac{1}{380} \approx 0.003$

(b) The teacher picks 2 students at random to hand out papers. What is the probability that Ana and Ben are picked?

No, it is the same job. ${}_{20}C_2 = 190$. $P = \frac{1}{190} \approx 0.005$

(c) Why is the answer to (b) 2 times the answer to (a)?

In (a), only 1 of the 2 orders of Ana and Ben works. In (b), both orders are the same pair.

(d) A playlist of 6 songs plays in a random order. What is the probability that the songs play in the order you like best?

Yes, order matters. $6! = 720$. $P = \frac{1}{720} \approx 0.001$

(e) A 4-digit locker code uses the digits 1 to 8 with no repeats. The code is chosen at random. What is the probability that the code is 1357?

Yes. 1357 and 7531 are different codes. ${}_8P_4 = 1{,}680$. $P = \frac{1}{1{,}680} \approx 0.0006$

(f) A student says, "3 of 10 people win the same prize. Order matters, so the probability that Ana, Ben, and Carla win is $\frac{1}{720}$." Is the student right? Explain.

No. It is the same prize, so order does not matter. ${}_{10}C_3 = 120$, so $P = \frac{1}{120}$.

Test-Style Questions

A test asks the same questions with more words. Read slowly. Circle the best answer.

15. In a group of 60 students, 28 are on a sports team, 22 are in a club, and 10 are both on a sports team and in a club. A student is selected at random from the group. What is the probability that the student is on a sports team or in a club?

    A $\frac{1}{6}$

    B $\frac{1}{2}$

    (C) $\frac{2}{3}$

    D $\frac{5}{6}$

    $\frac{28}{60} + \frac{22}{60} - \frac{10}{60} = \frac{40}{60} = \frac{2}{3}$

16. A school has two copy machines that work independently of each other. On any given day, the probability that the first machine is broken is 0.1, and the probability that the second machine is broken is 0.2. What is the probability that at least one of the machines works on a given day?

    A 0.02

    B 0.28

    C 0.72

    (D) 0.98

    $P(\text{both broken}) = 0.1 \cdot 0.2 = 0.02$. $1 - 0.02 = 0.98$

17. The two-way table shows the results of a survey of 90 students about their favorite type of movie. A student from the survey is selected at random. Given that the student is in 12th grade, what is the probability that the student prefers action movies?

	Action	**Comedy**	**Total**
11th grade	20	25	45
12th grade	30	15	45
Total	50	40	90

    A $\frac{1}{3}$

    B $\frac{5}{9}$

    C $\frac{3}{5}$

    (D) $\frac{2}{3}$

    Only the 45 students in 12th grade. $\frac{30}{45} = \frac{2}{3}$

18. The probability of rain is 30% on each of the next 5 days. A student wants to use a simulation to model whether it rains on one of these days. Which simulation can the student use?

(A) Pick a random digit from 0 to 9. The digits 0, 1, and 2 mean rain.
B Pick a random digit from 0 to 9. The digits 0, 1, 2, and 3 mean rain.
C Roll a number cube. The numbers 1 and 2 mean rain.
D Flip a coin. Heads means rain.

19. How many different 4-letter arrangements can be made from the letters in the word SPORT if no letter can be used more than once?

A 5
B 24
(C) 120
D 625

${}_5P_4 = 5 \cdot 4 \cdot 3 \cdot 2 = 120$

20. A student council has 12 members. A committee of 4 members will be selected from the council. How many different committees of 4 members are possible?

A 48
(B) 495
C 11,880
D 20,736

Order does not matter. ${}_{12}C_4 = 495$

21. A teacher will randomly select 2 students from a group of 4 juniors and 6 seniors. What is the probability that both students selected are juniors?

A $\frac{1}{15}$
(B) $\frac{2}{15}$
C $\frac{4}{25}$
D $\frac{2}{5}$

${}_4C_2 = 6$ and ${}_{10}C_2 = 45$. $\frac{6}{45} = \frac{2}{15}$